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Inventor(s)	Ramakrishnan; Senthil

Security management service for internet-of-things devices

Abstract

A security management service for Internet-of-things devices can obtain, from an Internet-of-things device and via a cellular network, a request by the Internet-of-things device to register with the cellular network, the Internet-of-things device including a universal integrated circuit card that stores a unique identifier for the Internet-of-things device and a cellular transceiver. The security management service can obtain behavioral data describing activity associated with the Internet-of-things device and can determine, based on the identity data and the behavioral data, if the Internet-of-things device is operating normally and as expected and if the Internet-of-things device is under the control of any unauthorized entity before allowing the Internet-of-things device to register with the cellular network.

Inventors:	Ramakrishnan; Senthil (Atlanta, GA)
Applicant:	AT&T Intellectual Property I, L.P. (Atlanta, GA)
Family ID:	90470308
Assignee:	AT&T Intellectual Property I, L.P. (Atlanta, GA)
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Primary Examiner: Sabouri; Mazda

Attorney, Agent or Firm: Hartman & Citrin LLC

Background/Summary

BACKGROUND

(1) Network Identity and authentication are often used in cellular networks. Cellular devices can be equipped with a subscriber identity module (“SIM”) or other form of universal integrated circuit card (“UICC”). The universal integrated circuit card can contain a unique identity that can be used to request and/or provide network resources such as voice connectivity, data connectivity, Internet access, etc. Because cellular-connected devices generally include some form of a universal integrated circuit card, identity of cellular devices and other security considerations often can be managed and/or tracked by cellular network providers.

(2) For non-cellular-enabled devices, security processes and/or technologies can vary widely. Some resources may require certificates to communicate, and various types of authentication and/or identity determination techniques are used to attempt to identify and/or authorize various types of communications associated with Internet-of-things devices. In some cases, proprietary authentication and/or identity technologies are used for Internet-of-things devices, which can complicate management networks and/or network resources.

SUMMARY

(3) The present disclosure is directed to security management service for Internet-of-things devices. One or more Internet-of-things devices can include a universal integrated circuit card. The universal integrated circuit card can store universally unique and immutable identity data associated with the Internet-of-things device, and the Internet-of-things device also can execute a security application that can be executed by the Internet-of-things device to access identity data, request registration with a cellular network, store and/or provide certificates, and/or to perform other functionality. The Internet-of-things device can request registration with the cellular network by sending registration data to a security management service, which can be hosted and/or executed by the server computer.

(4) The security management service can analyze or trigger analysis of the identity data and/or other data describing behavior of the Internet-of-things device to determine if the Internet-of-things device is to be registered with the cellular network. In particular, the security management service

can determine, based on behavior of the Internet-of-things device, the identity data, identifying information stored in the identity database, and information stored in the device inventory, if the Internet-of-things device is accurately identified (e.g., that the identity data matches the known identifying information associated with the Internet-of-things device), that the Internet-of-things device is operating normally and as expected, and that the Internet-of-things device is not currently under the control of any unauthorized entity and/or malware. If the security management service so determines, the security management service can register and/or trigger registration of the Internet-of-things device with the cellular network.

(5) At some time after registering with the cellular network, the Internet-of-things device can request access to a resource on the network such as one or more third party services, one or more components of network infrastructure, and/or one or more cloud applications. The security management service can determine if the access requested by the Internet-of-things device (e.g., by way of an implicit or explicit resource access request) is to be granted. In various embodiments, the security management service can again analyze or trigger analysis of the identity data and/or other data describing behavior of the Internet-of-things device to determine if the Internet-of-things device is to be allowed to access the resource as requested via the cellular network. The security management service can determine, based on behavior of the Internet-of-things device, the identity data, identifying information stored in the identity database, and information stored in the device inventory, if the Internet-of-things device is accurately identified (e.g., that the identity data matches the known identifying information associated with the Internet-of-things device), that the Internet-of-things device is operating normally and as expected, and that the Internet-of-things device is not currently under the control of any unauthorized entity and/or malware. If the security management service so determines, the security management service can allow the requested access to the resource.

(6) If the access is granted, the security management service also can determine (and/or invoke the credential manager to determine) if a certificate is required for communications between the Internet-of-things device and the resource. If the certificate is required for communications between the Internet-of-things device and the resource, the security management service can provide the certificate to the resource. The security management service also can determine (and/or invoke the routing controller to determine) a routing for session data between the Internet-of-things device and the resource. The routing can include a direct session between the Internet-of-things device and the resource or an indirect route between the Internet-of-things device via the server computer. These and other aspects of the concepts and technologies disclosed herein for a security management service for Internet-of-things devices will be illustrated and described in additional detail herein.

(7) According to one aspect of the concepts and technologies disclosed herein, a system is disclosed. The system can include a processor and a memory. The memory can store computer-executable instructions that, when executed by the processor, cause the processor to perform operations. The operations can include obtaining, from an Internet-of-things device and via a portion of a cellular network, registration data. Obtaining the registration data can correspond to a request by the Internet-of-things device to register with the cellular network, and the Internet-of-things device includes a universal integrated circuit card that stores a unique identifier for the Internet-of-things device and a cellular transceiver. The operations further can include obtaining, from the registration data, identity data that includes the unique identifier for the Internet-of-things device; obtaining behavioral data associated with the Internet-of-things device, the behavioral data describing activity associated with the Internet-of-things device; determining, based on the identity data and the behavioral data, if the Internet-of-things device is operating normally and as expected and if the Internet-of-things device is under the control of any unauthorized entity; and if a determination is made that the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under the control of any unauthorized entity, allowing the

Internet-of-things device to register with the cellular network.

(8) In some embodiments, the operations can further include receiving, from the Internet-of-things device and via the cellular network, another request to access a resource via another network; analyzing identity information associated with the Internet-of-things device and the behavioral data to determine if the access requested is to be granted; and if a determination is made that the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under the control of any unauthorized entity, allowing the Internet-of-things device to access the resource via the cellular network. In some embodiments, the operations can further include determining if the resource requires a certificate associated with the Internet-of-things device to allow the Internet-of-things device to access the resource; and if a determination is made that the resource requires the certificate, providing the certificate to the resource, wherein the certificate is obtained with the registration data and from the universal integrated circuit card.

(9) In some embodiments, the operations can further include determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing including a direct connection between the Internet-of-things device and the resource via the cellular network and the other network. In some embodiments, the operations can further include determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing including an indirect connection between the Internet-of-things device and the resource, the indirect connection including communications via the cellular network, a server computer that executes a security management service, and the other network. In some embodiments, the resource can include a cloud application. In some embodiments, the Internet-of-things device can store a certificate in the universal integrated circuit card at the Internet-of-things device, wherein the certificate is stored in a secure memory of the universal integrated circuit card.

(10) According to another aspect of the concepts and technologies disclosed herein, a method is disclosed. The method can include obtaining, at a server computer including a processor and from an Internet-of-things device and via a portion of a cellular network, registration data. Obtaining the registration data can correspond to a request by the Internet-of-things device to register with the cellular network, and the Internet-of-things device includes a universal integrated circuit card that stores a unique identifier for the Internet-of-things device and a cellular transceiver. The method further can include obtaining, by the processor and from the registration data, identity data that includes the unique identifier for the Internet-of-things device; obtaining, by the processor, behavioral data associated with the Internet-of-things device, the behavioral data describing activity associated with the Internet-of-things device; determining, by the processor and based on the identity data and the behavioral data, if the Internet-of-things device is operating normally and as expected and if the Internet-of-things device is under the control of any unauthorized entity; and if a determination is made that the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under the control of any unauthorized entity, allowing, by the processor, the Internet-of-things device to register with the cellular network.

(11) In some embodiments, the method can further include receiving, from the Internet-of-things device and via the cellular network, another request to access a resource via another network; analyzing identity information associated with the Internet-of-things device and the behavioral data to determine if the access requested is to be granted; and if a determination is made that the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under the control of any unauthorized entity, allowing the Internet-of-things device to access the resource via the cellular network. In some embodiments, the method can further include determining if the resource requires a certificate associated with the Internet-of-things device to allow the Internet-of-things device to access the resource; and if a determination is made that the resource requires the certificate, providing the certificate to the resource, wherein the certificate is obtained with the registration data and from the universal integrated circuit card.

(12) In some embodiments, the method can further include determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing including a direct connection between the Internet-of-things device and the resource via the cellular network and the other network. In some embodiments, the method can further include determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing including an indirect connection between the Internet-of-things device and the resource, the indirect connection including communications via the cellular network, a server computer that executes a security management service, and the other network. In some embodiments, the resource can include a cloud application. In some embodiments, the Internet-of-things device can store a certificate in the universal integrated circuit card at the Internet-of-things device, wherein the certificate is stored in a secure memory of the universal integrated circuit card.

(13) According to yet another aspect of the concepts and technologies disclosed herein, a computer storage medium is disclosed. The computer storage medium can store computer-executable instructions that, when executed by a processor, cause the processor to perform operations. The operations can include obtaining, from an Internet-of-things device and via a portion of a cellular network, registration data. Obtaining the registration data can correspond to a request by the Internet-of-things device to register with the cellular network, and the Internet-of-things device includes a universal integrated circuit card that stores a unique identifier for the Internet-of-things device and a cellular transceiver. The operations further can include obtaining, from the registration data, identity data that includes the unique identifier for the Internet-of-things device; obtaining behavioral data associated with the Internet-of-things device, the behavioral data describing activity associated with the Internet-of-things device; determining, based on the identity data and the behavioral data, if the Internet-of-things device is operating normally and as expected and if the Internet-of-things device is under the control of any unauthorized entity; and if a determination is made that the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under the control of any unauthorized entity, allowing the Internet-of-things device to register with the cellular network.

(14) In some embodiments, the operations can further include receiving, from the Internet-of-things device and via the cellular network, another request to access a resource via another network; analyzing identity information associated with the Internet-of-things device and the behavioral data to determine if the access requested is to be granted; and if a determination is made that the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under the control of any unauthorized entity, allowing the Internet-of-things device to access the resource via the cellular network. In some embodiments, the operations can further include determining if the resource requires a certificate associated with the Internet-of-things device to allow the Internet-of-things device to access the resource; and if a determination is made that the resource requires the certificate, providing the certificate to the resource, wherein the certificate is obtained with the registration data and from the universal integrated circuit card.

(15) In some embodiments, the operations can further include determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing including a direct connection between the Internet-of-things device and the resource via the cellular network and the other network. In some embodiments, the operations can further include determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing including an indirect connection between the Internet-of-things device and the resource, the indirect connection including communications via the cellular network, a server computer that executes a security management service, and the other network. In some embodiments, the resource can include a cloud application. In some embodiments, the Internet-of-things device can store a certificate in the universal integrated circuit card at the Internet-of-things device, wherein the certificate is stored in a secure memory of the universal integrated circuit card.

(16) Other systems, methods, and/or computer program products according to embodiments will be or become apparent to one with skill in the art upon review of the following drawings and detailed description. It is intended that all such additional systems, methods, and/or computer program products be included within this description and be within the scope of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a system diagram illustrating an illustrative operating environment for various embodiments of the concepts and technologies described herein.
- (2) FIG. 2 is a flow diagram showing aspects of a method for an Internet-of-things device registering with a security management service, according to an illustrative embodiment of the concepts and technologies described herein.
- (3) FIG. 3 is a flow diagram showing aspects of a method for registering an Internet-of-things device with a security management service, according to an illustrative embodiment of the concepts and technologies described herein.
- (4) FIG. 4 is a flow diagram showing aspects of a method for evaluating a resource access request from an Internet-of-things device using a security management service, according to an illustrative embodiment of the concepts and technologies described herein.
- (5) FIG. 5 schematically illustrates a network, according to an illustrative embodiment of the concepts and technologies described herein.
- (6) FIG. 6 is a block diagram illustrating an example computer system configured to implement various aspects of the concepts and technologies disclosed herein, according to some illustrative embodiments of the concepts and technologies described herein.
- (7) FIG. 7 is a block diagram illustrating an example Internet-of-things device configured to interact with a security management service, according to some illustrative embodiments of the concepts and technologies described herein.
- (8) FIG. 8 is a diagram illustrating a computing environment capable of implementing aspects of the concepts and technologies disclosed herein, according to some illustrative embodiments of the concepts and technologies described herein.

DETAILED DESCRIPTION

- (9) The following detailed description is directed to security management service for Internet-of-things devices. One or more Internet-of-things devices can include a universal integrated circuit card. The universal integrated circuit card can store universally unique and immutable identity data associated with the Internet-of-things device, and the Internet-of-things device also can execute a security application that can be executed by the Internet-of-things device to access identity data, request registration with a cellular network, store and/or provide certificates, and/or to perform other functionality. The Internet-of-things device can request registration with the cellular network by sending registration data to a security management service, which can be hosted and/or executed by the server computer.
- (10) The security management service can analyze or trigger analysis of the identity data and/or other data describing behavior of the Internet-of-things device to determine if the Internet-of-things device is to be registered with the cellular network. In particular, the security management service can determine, based on behavior of the Internet-of-things device, the identity data, identifying information stored in the identity database, and information stored in the device inventory, if the Internet-of-things device is accurately identified (e.g., that the identity data matches the known identifying information associated with the Internet-of-things device), that the Internet-of-things device is operating normally and as expected, and that the Internet-of-things device is not currently under the control of any unauthorized entity and/or malware. If the security management service so

determines, the security management service can register and/or trigger registration of the Internet-of-things device with the cellular network.

(11) At some time after registering with the cellular network, the Internet-of-things device can request access to a resource on the network such as one or more third party services, one or more components of network infrastructure, and/or one or more cloud applications. The security management service can determine if the access requested by the Internet-of-things device (e.g., by way of an implicit or explicit resource access request) is to be granted. In various embodiments, the security management service can again analyze or trigger analysis of the identity data and/or other data describing behavior of the Internet-of-things device to determine if the Internet-of-things device is to be allowed to access the resource as requested via the cellular network. The security management service can determine, based on behavior of the Internet-of-things device, the identity data, identifying information stored in the identity database, and information stored in the device inventory, if the Internet-of-things device is accurately identified (e.g., that the identity data matches the known identifying information associated with the Internet-of-things device), that the Internet-of-things device is operating normally and as expected, and that the Internet-of-things device is not currently under the control of any unauthorized entity and/or malware. If the security management service so determines, the security management service can allow the requested access to the resource.

(12) If the access is granted, the security management service also can determine (and/or invoke the credential manager to determine) if a certificate is required for communications between the Internet-of-things device and the resource. If the certificate is required for communications between the Internet-of-things device and the resource, the security management service can provide the certificate to the resource. The security management service also can determine (and/or invoke the routing controller to determine) a routing for session data between the Internet-of-things device and the resource. The routing can include a direct session between the Internet-of-things device and the resource or an indirect route between the Internet-of-things device via the server computer. These and other aspects of the concepts and technologies disclosed herein for a security management service for Internet-of-things devices will be illustrated and described in additional detail herein

(13) While the subject matter described herein is presented in the general context of program modules that execute in conjunction with the execution of an operating system and application programs on a computer system, those skilled in the art will recognize that other implementations may be performed in combination with other types of program modules. Generally, program modules include routines, programs, components, data structures, and other types of structures that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the subject matter described herein may be practiced with other computer system configurations, including hand-held devices, multiprocessor systems, microprocessor-based or programmable consumer electronics, minicomputers, mainframe computers, and the like.

(14) Referring now to FIG. 1, aspects of an operating environment **100** for various embodiments of the concepts and technologies disclosed herein for security management service for Internet-of-things devices will be described, according to an illustrative embodiment. The operating environment **100** shown in FIG. 1 includes a server computer **102**. The server computer **102** can operate in communication with and/or as part of a communications network (“network”) **104**, though this is not necessarily the case in all embodiments.

(15) According to various embodiments, the functionality of the server computer **102** may be provided by one or more server computers, desktop computers, mobile telephones, laptop computers, set-top boxes, other computing systems, and the like. It should be understood that the functionality of the server computer **102** may be provided by a single device, by two or more similar devices, and/or by two or more dissimilar devices. For purposes of describing the concepts and technologies disclosed herein, the server computer **102** is described herein as a server computer

such as an application server, a web server, or the like. It should be understood that this embodiment is illustrative, and should not be construed as being limiting in any way.

(16) The server computer **102** can execute an operating system (not labeled in FIG. **1**) and one or more application programs such as, for example, a security management service **106**, a credential manager **108**, and a routing controller **110**. These application programs can generate, store, and/or interact with an identity database **112**, a device inventory **114**, and/or other data sources as will be illustrated and described herein. The operating system can include a computer program that can control the operation of the server computer **102**. The application programs can include executable programs that can be configured to execute on top of the operating system to provide various functions as illustrated and described herein.

(17) Although the security management service **106**, the credential manager **108**, and the routing controller **110** are illustrated as components of the server computer **102**, it should be understood that each of these components, or combinations thereof, may be embodied as or in stand-alone devices or components thereof operating as part of or in communication with the network **104** and/or the server computer **102**. Similarly, while the device inventory **114** and the identity database **112** are illustrated as being stored in a data storage location (e.g., a memory) of the server computer **102**, it should be understood that these and other data can be stored in data repositories, data storage devices, databases, data stores, and/or other data storage resources associated with and/or accessible to the server computer **102**. As such, the illustrated embodiment should be understood as being illustrative of only some contemplated embodiments and should not be construed as being limiting in any way.

(18) The security management service **106** can be configured to manage communications of one or more Internet-of-things devices **116A-N** (hereinafter collectively and/or generically referred to as “Internet-of-things devices **116**”). One or more of the Internet-of-things devices **116** can execute a security application **118** and one or more identity modules such as a SIM, UICC, or the like (labeled in FIG. **1** and hereinafter referred to as the “universal integrated circuit card **120**”). The security application **118** of the Internet-of-things device **116** can be configured to collect contextual data, generate requests for registration, generate requests to access resources, to obtain and/or store credentials, and/or other functionality of the Internet-of-things device **116** as illustrated and described herein. According to various embodiments of the concepts and technologies disclosed herein, Internet-of-things devices **116** for which security is managed by the security management service **106** can include the universal integrated circuit card **120**. According to various embodiments, the universal integrated circuit card **120** of the Internet-of-things devices **116** can store one or more certificates **122** or other identity and/or authentication technologies (e.g., tokens, or the like). It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(19) According to various embodiments of the concepts and technologies disclosed herein, the server computer **102** and the Internet-of-things devices **116** can be configured for wireless communications via a cellular network **124**. Thus, although not illustrated in FIG. **1**, the Internet-of-things devices **116** also can include wireless transceivers. An example architecture for some embodiments of the Internet-of-things devices **116** is illustrated and described below with reference to FIG. **7**.

(20) According to various embodiments of the concepts and technologies disclosed herein, the security management service **106** can be configured to authenticate the Internet-of-things devices **116** and/or manage communications associated with the Internet-of-things devices **116**. In particular, the Internet-of-things devices **116** can be configured to register for communications via the cellular network **124**. In particular, the Internet-of-things devices **116** can be configured to request registration with the cellular network **124** at various times such as on power up, when a transceiver is activated, at other times, or the like. According to various embodiments, the Internet-of-things devices **116** can request the registration via exchanging one or more instances of

registration data **126** with the server computer **102**. Because registration can be requested in additional and/or alternative manners, it should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(21) The registration data **126** generated by the Internet-of-things device **116** can include, in some embodiments, a request to access the cellular network **124** (e.g., a ping to a cellular tower requesting access, or the like). In some instances, the server computer **102** can be configured to obtain or receive the registration data **126** and determine, based on the registration data **126** and/or other information, if the Internet-of-things device **116** is to be registered with the cellular network **124**. In some embodiments, the registration data **126** can include identity data, one or more certificates **122**, and/or other data. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(22) The identity data can include a unique and/or immutable identifier for the Internet-of-things device **116**. The identity data therefore can correspond to a string of characters that can uniquely identify a single device and, in theory, a user associated with the device. Thus, the identity data illustrated and described herein can include any format of unique identifier such as a globally unique identifier (“GUID”) or other string or object that can uniquely identify one, and only one, of the Internet-of-things devices **116**. As such, the security management service **106** can know, based on the identity data included in the registration data **126**, exactly what device (e.g., one of the Internet-of-things devices **116**) has generated the registration data **126**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(23) In various embodiments of the concepts and technologies disclosed herein, the universal integrated circuit card **120** can also store one or more certificates **122**. Thus, for example, a certificate **122** can be stored locally at one of the Internet-of-things devices **116**, for example, in a secure memory portion of the universal integrated circuit card **120**. In various embodiments, a user or other entity can load a certificate **122** to the universal integrated circuit card **120** for use at various times and/or for various purposes as will be illustrated and described herein. Additionally, or alternatively, the universal integrated circuit card **120** can be preloaded or pre-flashed with a certificate **122** for use in association with secure resources. Thus, as will be explained in more detail herein, the security management service **106** can be configured to obtain and store the certificate **122** associated with a particular Internet-of-things device **116** and to store the certificate **122** to provide to resources that request a certificate **122**. In some other embodiments, the security management service **106** can be configured to request the certificate **122** from the Internet-of-things device **116** when a communication requires the certificate **122**. Thus, it can be appreciated that the server computer **102** can store or obtain one or more certificates **122**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(24) The other data can include other information associated with the Internet-of-things device **116** such as, for example, contextual information that describes how the Internet-of-things device **116** is being used and/or has been used in the past. Thus, the other data can be used by the security management service **106** to evaluate behavior of the Internet-of-things device **116** to determine if the Internet-of-things device **116** is operating as expected and/or normally, if a requested registration or communication is expected, if any malware or unauthorized control of the Internet-of-things device **116** is detected, for other reasons, or the like. Thus, the registration data **126** can include information that may be used to authorize (or not authorize) registration and/or communications associated with the Internet-of-things device **116** via the cellular network **124**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(25) The security management service **106** can be configured to detect the request to register with the cellular network **124** (e.g., via receipt of the registration data **126**) and to determine if the Internet-of-things device **116** is to be registered with the cellular network **124**. In particular, the

security management service **106** can be configured to access and analyze, or to invoke the credential manager **108** to access and analyze, the registration data **126** (e.g., the identity data and the other data), and can obtain and analyze the device inventory **114** and/or identity database **112** to determine if the identity associated with the Internet-of-things device **116** is authorized to register with the cellular network **124**.

(26) The security management service **106** also can be configured to determine, based on performing an anomaly detection procedure and/or executing an anomaly detection module of the security management service **106**, if the Internet-of-things device **116** is operating normally and/or as expected. The security management service **106** also can determine, e.g., based on performing threat intelligence and/or executing a threat intelligence module of the security management service **106**, if the Internet-of-things device **116** is infected and/or otherwise under the control of any unauthorized entity or process. Based on these and/or other considerations as illustrated and described herein, the security management service **106** can determine if the Internet-of-things device **116** is to be registered with the cellular network **124**.

(27) In particular, if the security management service **106** determines that the identity data (e.g., the unique identity stored in the universal integrated circuit card **120**) associated with the Internet-of-things device **116** is authorized to register with the cellular network **124**; that the corroborating identity information matches the identity data obtained from the universal integrated circuit card **120** of the Internet-of-things device **116**; that the Internet-of-things device **116** is operating as expected and without anomalies; and that no malware and/or unauthorized control of the Internet-of-things device **116** is detected; the Internet-of-things device **116** can be determined to be entitled to registration with the cellular network **124**. If the security management service **106** determines, on the other hand, that the identity data (e.g., the unique identity stored in the universal integrated circuit card **120**) associated with the Internet-of-things device **116** is not authorized to register with the cellular network **124**; that the corroborating identity information does not match the identity data obtained from the universal integrated circuit card **120** of the Internet-of-things device **116**; that the Internet-of-things device **116** is not operating as expected and/or is operating with one or more anomaly; or that malware or unauthorized control of the Internet-of-things device **116** is detected; the Internet-of-things device **116** can be determined not to be entitled to registration with the cellular network **124**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(28) If the security management service **106** determines that the Internet-of-things device **116** is not to be registered, the security management service **106** can block or not allow registration of the Internet-of-things device **116** and/or deny provisioning of services on the cellular network **124** to the Internet-of-things device **116**. As such, it can be appreciated that in some embodiments, the security management service **106** can take action to prevent registration of the Internet-of-things device **116** on the cellular network **124**, while in some other embodiments, the security management service **106** may refrain from taking action to register the Internet-of-things device **116** on the cellular network **124** (without actually taking any action). Because registration of the Internet-of-things device **116** can be blocked or not granted in additional and/or alternative manners, it should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way. If the security management service **106** determines that the Internet-of-things device **116** is to be registered, the security management service **106** can register the Internet-of-things device **116** on the cellular network **124** and/or allow the Internet-of-things device **116** to register with the cellular network **124**. Because the registration with the cellular network **124** can be enabled and/or allowed in additional and/or alternative manners, it should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(29) According to various embodiments, the security management service **106** also can be configured to communicate with one or more resources via the network **104**. For example, the

security management service **106** can be configured to access one or more third party services **128** that can operate on or in communication with the network **104**. The third party services **128** can include, for example, one or more web servers, web sites, file servers, websites, combinations thereof, or the like.

(30) The security management service **106** also can be configured to access network infrastructure **130** associated with the network **104**. The network infrastructure **130** can include servers, databases, applications, or the like, operating on and/or in communication with the network **104**. Additionally, the security management service **106** also can be configured to access one or more cloud applications **132**, which can operate on and/or in communication with the network **104**. It can be appreciated that the cloud applications **132** can include one or more applications or services provided by one or more distributed computing devices or environments (e.g., virtual machines, data centers, and/or other processing and/or data storage resources).

(31) It can be appreciated that the cloud applications **132** can be similar to, include, and/or be included in, the third party services **128** and/or network infrastructure **130**, in some embodiments. As such, the third party services **128**, the network infrastructure **130**, and the cloud applications **132** are collectively and/or generically referred to herein at times as “resources” and/or a “resource.” Thus, a “resource” as referred to herein can include one or more of the third party services **128**, one or more component of the network infrastructure **130**, and/or one or more of the cloud applications **132**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(32) At some point in time after registering with the cellular network **124**, the Internet-of-things device **116** can request access to one of the resources, for example by way of a resource access request **134**. The resource access request **134** can correspond to an explicit request to access one of the resources, in some embodiments, and can be generated by the Internet-of-things device **116**. In some other embodiments, the resource access request **134** can correspond to an attempt to connect to one of the resources such as, for example, an attempt to access one or more of the third party services **128**, an attempt to access one or more component of the network infrastructure **130**, and/or an attempt to access and/or call one or more of the cloud applications **132**. Thus, it can be appreciated that the resource access request **134** can correspond to an implicit or explicit request to access one or more of the resources.

(33) The security management service **106** can be configured to determine if the access requested by way of the resource access request **134** is to be granted, not granted, allowed, blocked, or the like. In particular, the security management service **106** can analyze (e.g., by executing the credential manager **108**) the resource access request **134**, the access requested, the identity database **112**, the device inventory **114**, and/or activity fingerprints (e.g., included in the registration data **126**). If the security management service **106** determines that the identity data (e.g., the unique identity stored in the universal integrated circuit card **120**) associated with the Internet-of-things device **116** is the same entity that was authorized to register with the cellular network **124**; that the corroborating identity information matches the identity data obtained from the universal integrated circuit card **120** of the Internet-of-things device **116**; that the Internet-of-things device **116** is operating as expected and without anomalies; and that no malware and/or unauthorized control of the Internet-of-things device **116** is detected; the Internet-of-things device **116** can be determined to be entitled to access the requested resource and/or to obtain the requested access via the cellular network **124**.

(34) If the security management service **106** determines that the identity data (e.g., the unique identity stored in the universal integrated circuit card **120**) associated with the Internet-of-things device **116** is not authorized to register with the cellular network **124**; that the corroborating identity information does not match the identity data obtained from the universal integrated circuit card **120** of the Internet-of-things device **116**; that the Internet-of-things device **116** is not operating as expected and/or is operating with one or more anomaly; or that malware or unauthorized control

of the Internet-of-things device **116** is detected; the Internet-of-things device **116** can be determined not to be entitled to access the resource via the cellular network **124**.

(35) If the security management service **106** approves the access to the resource by the Internet-of-things device **116**, the security management service **106** can determine if a certificate **122** is required for the requested access. For example, if the resource being accessed requires a certificate **122** to establish an encrypted and/or verified session with the Internet-of-things device **116**, the security management service **106** can determine that a certificate **122** is required. If the security management service **106** determines that the certificate **122** is required, the security management service **106** can provide a copy of the certificate **122** to the resource. For example, as shown in FIG. **1**, if the Internet-of-things device **116** is approved to access a cloud application **132**, the security management service **106** can be configured to provide, to the cloud application **132**, the certificate **122** of the Internet-of-things device **116**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(36) Additionally, the security management service **106** can determine, or can invoke the routing controller **110** to determine, for approved access to a resource, a routing to be used for the access to the resource. As used herein, the “routing” for the access can include one or more data paths for a session associated with the requested access. According to various embodiments, the determined routing can include direct communications between the Internet-of-things device **116** and the requested resource (e.g., exchange of data associated with the session, hereinafter referred to as “session data **136**”) via the cellular network **124** and the network **104**. In some other embodiments, the determined routing can include indirect communications between the Internet-of-things device **116** and the requested resource (e.g., exchange of the session data **136** via the server computer **102** (which can manage the communications), the cellular network **124**, and the network **104**). In some embodiments, the server computer **102** can function as an intermediary to manage security at each communication (e.g., exchange of session data **136**) between the Internet-of-things device **116** and the resource. In some other embodiments, direct communication between the Internet-of-things device **116** and the resource without the server computer **102** being an intermediary can reduce latency and/or otherwise improve quality of service. Because a direct or indirect routing can be selected for additional and/or alternative reasons, and because additional and/or alternative benefits can be realized by either approach, it should be understood that the above examples are illustrative and should not be construed as being limiting in any way.

(37) In practice, one or more Internet-of-things devices **116** can include a universal integrated circuit card **120**. The universal integrated circuit card **120** can store universally unique and immutable identity data associated with the Internet-of-things device **116**, and the Internet-of-things device **116** also can execute a security application **118** for accessing identity data, requesting registration with a cellular network **124**, storing and/or providing certificates **122** and/or other functionality. The Internet-of-things device **116** can request registration with the cellular network **124** by sending registration data **126** to a security management service **106**, which can be hosted and/or executed by the server computer **102**.

(38) The security management service **106** can analyze or trigger analysis of the identity data and/or other data describing behavior of the Internet-of-things device **116** to determine if the Internet-of-things device **116** is to be registered with the cellular network **124**. In particular, the security management service **106** can determine, based on behavior of the Internet-of-things device **116**, the identity data, identifying information stored in the identity database **112**, and information stored in the device inventory **114**, if the Internet-of-things device **116** is accurately identified (e.g., that the identity data matches the known identifying information associated with the Internet-of-things device **116**), that the Internet-of-things device **116** is operating normally and as expected, and that the Internet-of-things device **116** is not currently under the control of any unauthorized entity and/or malware. If the security management service **106** so determines, the security management service **106** can register and/or trigger registration of the Internet-of-things device **116**

with the cellular network **124**.

(39) At some time after registering with the cellular network **124**, the Internet-of-things device **116** can request access to a resource on the network **104** such as one or more third party services **128**, one or more components of network infrastructure **130**, and/or one or more cloud applications. The security management service **106** can determine if the access requested by the Internet-of-things device **116** (e.g., by way of an implicit or explicit resource access request **134**) is to be granted. In various embodiments, the security management service **106** can again analyze or trigger analysis of the identity data and/or other data describing behavior of the Internet-of-things device **116** to determine if the Internet-of-things device **116** is to be allowed to access the resource as requested via the cellular network **124**. The security management service **106** can determine, based on behavior of the Internet-of-things device **116**, the identity data, identifying information stored in the identity database **112**, and information stored in the device inventory **114**, if the Internet-of-things device **116** is accurately identified (e.g., that the identity data matches the known identifying information associated with the Internet-of-things device **116**), that the Internet-of-things device **116** is operating normally and as expected, and that the Internet-of-things device **116** is not currently under the control of any unauthorized entity and/or malware. If the security management service **106** so determines, the security management service **106** can allow the requested access to the resource.

(40) If the access is granted, the security management service **106** also can determine (and/or invoke the credential manager **108** to determine) if a certificate **122** is required for communications between the Internet-of-things device **116** and the resource. If the certificate **122** is required for communications between the Internet-of-things device **116** and the resource, the security management service **106** can provide the certificate **122** to the resource. The security management service **106** also can determine (and/or invoke the routing controller **110** to determine) a routing for session data **136** between the Internet-of-things device **116** and the resource. The routing can include a direct session between the Internet-of-things device **116** and the resource or an indirect route between the Internet-of-things device **116** via the server computer **102**. These and other aspects of the concepts and technologies disclosed herein for a security management service **106** for Internet-of-things devices **116** will be illustrated and described in additional detail herein.

(41) It can be appreciated that the Internet-of-things device **116** can send the registration data **126** to the server computer **102** via the cellular network **124** prior to registration with the cellular network **124**. Namely, the Internet-of-things device **116** can request attachment to the cellular network **124**, in some embodiments, by way of sending the registration data **126** to the server computer **102** via a portion of the cellular network **124** (e.g., a tower, or the like), and the registration data **126** can be routed to the server computer **102** to determine if the Internet-of-things device **116** is to be registered with the cellular network **124** to enable the Internet-of-things device **116** to communicate in any other way (e.g., other than requesting access to the cellular network **124**) via the cellular network **124**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(42) FIG. 1 illustrates one server computer **102**, one network **104**, multiple Internet-of-things devices **116**, one cellular network **124**, one instance of third party services **128**, one instance of network infrastructure **130**, and one instance of cloud applications **132**. It should be understood, however, that various implementations of the operating environment **100** can include one or more than one server computer **102**; zero, one, or more than one network **104**; one or more than one Internet-of-things devices **116**; one or more than one cellular network **124**; zero, one, or more than one instance of third party services **128**; zero, one, or more than one instance of network infrastructure **130**; and/or zero, one, or more than one instance of cloud applications **132**. As such, the illustrated embodiment should be understood as being illustrative, and should not be construed as being limiting in any way.

(43) Turning now to FIG. 2, aspects of a method **200** for an Internet-of-things device registering

with a security management service will be described in detail, according to an illustrative embodiment. It should be understood that the operations of the methods disclosed herein are not necessarily presented in any particular order and that performance of some or all of the operations in an alternative order(s) is possible and is contemplated. The operations have been presented in the demonstrated order for ease of description and illustration. Operations may be added, omitted, and/or performed simultaneously, without departing from the scope of the concepts and technologies disclosed herein.

(44) It also should be understood that the methods disclosed herein can be ended at any time and need not be performed in its entirety. Some or all operations of the methods, and/or substantially equivalent operations, can be performed by execution of computer-readable instructions included on a computer storage media, as defined herein. The term “computer-readable instructions,” and variants thereof, as used herein, is used expansively to include routines, applications, application modules, program modules, programs, components, data structures, algorithms, and the like. Computer-readable instructions can be implemented on various system configurations including single-processor or multiprocessor systems, minicomputers, mainframe computers, personal computers, hand-held computing devices, microprocessor-based, programmable consumer electronics, combinations thereof, and the like.

(45) Thus, it should be appreciated that the logical operations described herein are implemented (1) as a sequence of computer implemented acts or program modules running on a computing system and/or (2) as interconnected machine logic circuits or circuit modules within the computing system. The implementation is a matter of choice dependent on the performance and other requirements of the computing system. Accordingly, the logical operations described herein are referred to variously as states, operations, structural devices, acts, or modules. These states, operations, structural devices, acts, and modules may be implemented in software, in firmware, in special purpose digital logic, and any combination thereof. As used herein, the phrase “cause a processor to perform operations” and variants thereof is used to refer to causing a processor of a computing system or device, such as the server computer **102** and/or one or more of the Internet-of-things devices **116**, to perform one or more operations and/or causing the processor to direct other components of the computing system or device to perform one or more of the operations.

(46) For purposes of illustrating and describing the concepts of the present disclosure, the method **200** is described herein as being performed by the Internet-of-things device **116** via execution of one or more software modules such as, for example, the security application **118** and/or one or more applications that can be implemented on the universal integrated circuit card **120**. It should be understood that additional and/or alternative devices and/or network nodes can provide the functionality described herein via execution of one or more modules, applications, and/or other software including, but not limited to, the security application **118**. Thus, the illustrated embodiments are illustrative, and should not be viewed as being limiting in any way.

(47) The method **200** begins at operation **202**. At operation **202**, the Internet-of-things device **116** can detect a registration request. According to various embodiments of the concepts and technologies disclosed herein, the Internet-of-things device **116** can be configured to register with a cellular network **124** when activated (e.g., the first time the Internet-of-things device **116** is turned on, each time the Internet-of-things device **116** is turned on and lacks access to any cellular network **124**, etc.), upon receiving a request to register with a cellular network **124** (e.g., via a hardware or software control for registering with the cellular network **124**), and/or via other commands. In some embodiments, an Internet-of-things device **116** may be interfaced with by a device such as a gateway, smartphone, or the like, and the request to register the Internet-of-things device **116** with the cellular network **124** may therefore be generated by an application, a user, or the like. Because the request to register the Internet-of-things device **116** with the cellular network **124** can be detected in numerous manners, it should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way.

(48) From operation **202**, the method **200** can proceed to operation **204**. At operation **204**, the Internet-of-things device **116** can obtain identity data from the universal integrated circuit card **120**. According to various embodiments of the concepts and technologies disclosed herein, the universal integrated circuit card **120** of the Internet-of-things device **116** can store identity information that can uniquely identify the Internet-of-things device **116**. As can be appreciated, a unique identifier such as a universal integrated circuit card **120** of the Internet-of-things device **116** can enable device-level control of access to ensure that only authorized Internet-of-things devices **116** are allowed to attach to the cellular network **124** and/or other entities via the cellular network **124**. Other attempts to identify Internet-of-things devices **116** (e.g., IP addresses, or the like) are easily replicated and/or cloned, while a unique identifier such as an identity associated with a universal integrated circuit card **120** can be known by the cellular network **124** (and/or entities thereon and/or associated therewith such as the server computer **102**). It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(49) From operation **204**, the method **200** can proceed to operation **206**. At operation **206**, the Internet-of-things device **116** can determine if a certificate **122** is stored in the universal integrated circuit card **120**. According to various embodiments of the concepts and technologies disclosed herein, the universal integrated circuit card **120** can be configured to store a certificate **122**. In some embodiments, the certificate **122** can be generated and stored in the universal integrated circuit card **120** when the universal integrated circuit card **120** is created, while in other embodiments, the certificate **122** can be obtained (e.g., via issuance by a certificate authority) and loaded to the universal integrated circuit card **120** by an authorized entity (e.g., an owner of the Internet-of-things device **116**, a gateway, or the like). Because the certificate **122** can be loaded to the universal integrated circuit card **120** in additional and/or alternative ways, it should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way.

(50) If the Internet-of-things device **116** determines, in operation **206**, that a certificate **122** is stored in the universal integrated circuit card **120**, the method **200** can proceed to operation **208**. At operation **208**, the Internet-of-things device **116** can obtain the certificate **122** from the universal integrated circuit card **120**. According to various embodiments of the concepts and technologies disclosed herein, the Internet-of-things device **116** can retrieve the certificate **122** from the universal integrated circuit card **120** wherein the universal integrated circuit card **120** can function as a data storage device for the certificate **122**.

(51) In some embodiments, the certificate **122** is stored in a secure memory that can be integrated into the universal integrated circuit card **120**. In some embodiments, the secure memory of the universal integrated circuit card **120** can be created when the universal integrated circuit card **120** is manufactured and/or loaded (e.g., “flashed”) by an authorized entity. As such, it can be appreciated that the certificate **122** can be encrypted and/or otherwise stored in a manner that is configured to allow only the Internet-of-things device **116** to retrieve and use the certificate **122** from the universal integrated circuit card **120**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(52) From operation **208**, the method **200** can proceed to operation **210**. The method **200** also can proceed to operation **210** if the Internet-of-things device **116** determines, in operation **206**, that a certificate **122** is not stored in the universal integrated circuit card **120**. At operation **210**, the Internet-of-things device **116** can register with a security management service **106**. According to various embodiments, the Internet-of-things device **116** can provide, to the server computer **102** or other device that hosts and/or executes the security management service **106**, registration data **126**.

(53) The registration data **126** can include the identity data obtained from the universal integrated circuit card **120** and, if included, the certificate **122** retrieved from the universal integrated circuit card **120**. The registration data **126** also can include other data as illustrated and described herein such as contextual information that indicates how the Internet-of-things device **116** is being used, geolocation data that identifies a geographic location of the Internet-of-things device **116**, and/or

other information. These and/or other data can be provided to the server computer **102** as a file or other data structure generated by the Internet-of-things device **116**.

(54) The security management service **106** can analyze the registration data **126** and, if appropriate, register the Internet-of-things device **116** for communications via the cellular network **124**. Thus, operation **210** can correspond to the Internet-of-things device **116** requesting provisioning of cellular network connectivity and the cellular network **124** granting cellular network connectivity. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(55) From operation **210**, the method **200** can proceed to operation **212**. At operation **212**, the Internet-of-things device **116** can request access to a resource such as the third party services **128**, the network infrastructure **130**, and/or the cloud applications **132**. It can be appreciated that the functionality of operation **212** can be performed at any time after registering with the cellular network **124** and is not necessarily immediately after registration.

(56) At any rate, in operation **212**, the Internet-of-things device **116** can request connection to one or more resources and/or the use of one or more resources. In some embodiments, for example, the Internet-of-things device **116** can request access to another network **104** and/or network infrastructure **130** associated therewith for various reasons; access to third party services **128**; and/or access to one or more cloud applications **132**. As such, operation **212** can correspond to the Internet-of-things device **116** attempting to access one or more resource and/or the Internet-of-things device **116** requesting access to the one or more resource. It should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way.

(57) From operation **212**, the method **200** can proceed to operation **214**. At operation **214**, the Internet-of-things device **116** can determine if the access requested in operation **212** has been granted by the security management service **106**. According to various embodiments of the concepts and technologies disclosed herein, the security management service **106** can determine if the Internet-of-things device **116** is to be allowed to access the requested resource, as will be explained in more detail herein, particularly with reference to FIG. 4, and grant or deny the access.

(58) If the Internet-of-things device **116** determines, in operation **214**, that the requested access has been granted, the Internet-of-things device **116** can exchange session data **136** with an entity. In particular, the Internet-of-things device **116** can be configured to communicate via the server computer **102**. In some embodiments, the server computer **102** can be configured (e.g., by execution of the security management service **106**) to send and receive the session data **136** as an intermediary and/or trusted entity for the Internet-of-things device **116** and/or the resources, thereby enhancing security of the communications. In some other embodiments, the server computer **102** can be configured (e.g., by execution of the security management service **106**) to enable communications directly between the Internet-of-things device **116** and the resource via direct exchange of session data **136** between the Internet-of-things device **116** and the resource. Thus, operation **214** can correspond to the server computer **102** enabling communications between the Internet-of-things device **116** and the resource (e.g., by either acting as an intermediary for the session data **136** or by allowing the devices to connect to one another). It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(59) If the Internet-of-things device **116** determines, in operation **214**, that the requested access has not been granted, method **200** can proceed to operation **218**. The method **200** also can proceed to operation **218** after the Internet-of-things device **116** exchanges session data **136** with the entity as illustrated in operation **216**. The method **200** can end at operation **218**.

(60) Turning now to FIG. 3, aspects of a method **300** for registering an Internet-of-things device with a security management service will be described in detail, according to an illustrative embodiment. For purposes of illustrating and describing the concepts of the present disclosure, the method **300** is described herein as being performed by the server computer **102** via execution of one or more software modules such as, for example, the security management service **106**. It

should be understood that additional and/or alternative devices and/or network nodes can provide the functionality described herein via execution of one or more modules, applications, and/or other software including, but not limited to, the security management service **106**. Thus, the illustrated embodiments are illustrative, and should not be viewed as being limiting in any way.

(61) The method **300** begins at operation **302**. At operation **302**, the server computer **102** can obtain registration data **126** from an Internet-of-things device **116**. As noted above, the registration data **126** can include identity data, a certificate **122**, other data (if included), and/or other information. In various embodiments of the concepts and technologies disclosed herein, the registration data **126** includes at least the identity data, wherein the identity data includes a unique identifier for the Internet-of-things device **116** (e.g., a unique identifier generated by the universal integrated circuit card **120**). In some other embodiments, the registration data **126** also can include a certificate **122** associated with the Internet-of-things device **116**. In yet other embodiments, the registration data **126** can include other information such as geolocation data, contextual data, or the like. Thus, operation **302** can include the server computer **102** obtaining at least a unique identifier associated with the Internet-of-things device **116**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(62) From operation **302**, the method **300** can proceed to operation **304**. At operation **304**, the server computer **102** can obtain data associated with the Internet-of-things device **116**. In some embodiments, for example, the server computer **102** can obtain inventory data associated with the Internet-of-things device **116**, identity data associated with the Internet-of-things device **116**, anomaly detection data associated with the Internet-of-things device **116**, threat intelligence data associated with the Internet-of-things device **116**, and/or other data.

(63) In some embodiments, operation **304** can include the server computer **102** obtaining inventory data from the device inventory **114**. Operation **304** also can include the server computer **102** obtaining the identity data from the identity database **112**. Operation **304** also can include the server computer **102** determining, based on anomaly detection and/or threat intelligence, if the Internet-of-things device **116** is healthy (e.g., not infected by malware and/or operating normally) or unhealthy and should or should not be registered with the cellular network **124**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(64) According to various embodiments, the device inventory **114** can correspond to a subscriber database that can link one or more universal integrated circuit cards (e.g., the universal integrated circuit card **120**) to one or more device details (e.g., manufacturer, model, serial number, etc.) and/or customer information associated with the Internet-of-things device **116** (e.g., a user or username associated with the Internet-of-things device **116**). Thus, the device inventory **114** can identify the Internet-of-things device **116** and/or one or more users associated with the Internet-of-things device **116**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(65) In various embodiments, the identity database **112** can include various other types of information and/or data that can identify the Internet-of-things device **116** (in addition to the identity data obtained from the universal integrated circuit card **120**) and can be used to correlate and/or corroborate the identity data. Thus, for example, the identity database **112** can store one or more IP addresses (e.g., a IPv4 address, an IPv6 address, etc.) for the Internet-of-things device **116**, a media access control (“MAC”) address for the Internet-of-things device **116**, access point names (“APNs”) and/or data network names (“DNNs”), location area codes (“LACs”) and/or tracking area codes (“TACs”), other identifying information, combinations thereof, or the like. These and/or other network attributes can be correlated to the Internet-of-things device **116** and used as one or more factors when authenticating (or not authenticating) the Internet-of-things device **116**. Thus, operation **304** can correspond to the server computer **102** obtaining these and/or other data for use in determining if the Internet-of-things device **116** is to be registered with the cellular network **124**.

Because other information can be used to determine if the Internet-of-things device **116** is to be registered with the cellular network **124**, it should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way.

(66) According to some embodiments, the anomaly detection (e.g., an anomaly detection module or process of the security management service **106**) can be configured to monitor and build a baseline for various Internet-of-things devices **116** (e.g., to establish “normal” behavior and “abnormal” behavior for Internet-of-things devices **116** when connected to the cellular network **124**. According to various embodiments, these baselines are determined using machine learning (“ML”) and/or artificial intelligence (“AI”) to analyze activity of Internet-of-things devices **116** that are known to be healthy and/or operating normally. With this baseline, activity and/or contextual data associated with Internet-of-things devices **116** can be analyzed to determine if the Internet-of-things devices **116** are healthy and/or operating normally. For example, the baseline can be correlated to activity and/or artefacts (e.g., the certificates **122**, keys, tokens, multi-factor authentication (“MFA”) attributes, etc.) to evaluate any Internet-of-things device **116** (for which a baseline exists) to determine if the Internet-of-things device **116** is operating normally and/or as expected. This correlation can be performed as part of operation **304**, in some embodiments.

(67) According to some embodiments, threat intelligence (e.g., a threat intelligence module or process of the security management service **106**) can be configured to analyze the Internet-of-things device **116** to detect any malware signatures and/or other activity that indicates that the Internet-of-things device **116** is infected and/or under the control of malware and/or other unauthorized entities. Thus, operation **304** can also include scanning the Internet-of-things device **116** behavior and/or communications to identify any infections of the Internet-of-things device **116**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(68) From operation **304**, the method **300** can proceed to operation **306**. At operation **306**, the server computer **102** can determine if the Internet-of-things device **116** is to be registered. Operation **306** can include considering the data and/or analysis obtained in operation **304** to determine if the Internet-of-things device **116** is to be registered with the cellular network **124**. Thus, it can be appreciated that operation **306** can include analyzing the device inventory **114** and/or identity database **112** to determine if the identity associated with the Internet-of-things device **116** is authorized to register with the cellular network **124**. Operation **306** also can include the server computer **102** determining, based on the anomaly detection, if the Internet-of-things device **116** is operating normally and/or as expected. Operation **306** also can include the server computer **102** determining, based on the threat intelligence, if the Internet-of-things device **116** is infected and/or otherwise under the control of any unauthorized entity or process.

(69) According to various embodiments of the concepts and technologies disclosed herein, if the server computer **102** determines that the identity data (e.g., the unique identity stored in the universal integrated circuit card **120**) associated with the Internet-of-things device **116** is authorized to register with the cellular network **124**; that the corroborating identity information matches the identity data obtained from the universal integrated circuit card **120** of the Internet-of-things device **116**; that the Internet-of-things device **116** is operating as expected and without anomalies; and that no malware and/or unauthorized control of the Internet-of-things device **116** is detected; the Internet-of-things device **116** can be determined to be entitled to registration with the cellular network **124**.

(70) Conversely, according to various embodiments of the concepts and technologies disclosed herein, if the server computer **102** determines that the identity data (e.g., the unique identity stored in the universal integrated circuit card **120**) associated with the Internet-of-things device **116** is not authorized to register with the cellular network **124**; that the corroborating identity information does not match the identity data obtained from the universal integrated circuit card **120** of the Internet-of-things device **116**; that the Internet-of-things device **116** is not operating as expected

and/or is operating with one or more anomaly; or that malware or unauthorized control of the Internet-of-things device **116** is detected; the Internet-of-things device **116** can be determined not to be entitled to registration with the cellular network **124**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(71) If the server computer **102** determines, in operation **306**, that the Internet-of-things device **116** is not to be registered, the method **300** can proceed to operation **308**. At operation **308**, the server computer **102** can block registration of the Internet-of-things device **116** and/or deny provisioning of services on the cellular network **124** to the Internet-of-things device **116**. As such, it can be appreciated that in some embodiments, the server computer **102** can take action to prevent registration of the Internet-of-things device **116** on the cellular network **124**, while in some other embodiments, the server computer **102** may refrain from taking action to register the Internet-of-things device **116** on the cellular network **124** (without actually taking any action). Because registration of the Internet-of-things device **116** can be blocked or not granted in additional and/or alternative manners, it should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way.

(72) If the server computer **102** determines, in operation **306**, that the Internet-of-things device **116** is to be registered, the method **300** can proceed to operation **310**. At operation **310**, the server computer **102** can register the Internet-of-things device **116** on the cellular network **124**. According to various embodiments of the concepts and technologies disclosed herein, the server computer **102** can provision the Internet-of-things device **116** on the cellular network **124** and/or trigger provisioning of service to the Internet-of-things device **116** on the cellular network **124**. Thus, operation **310** can correspond to the cellular network **124** enabling one or more connections between one or more devices of the cellular network **124** (e.g., a tower, or the like) and the Internet-of-things device **116** to enable the Internet-of-things device **116** to access the network **104** via the cellular network **124**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(73) From operation **310**, the method **300** can proceed to operation **312**. The method **300** can end at operation **312**.

(74) Turning now to FIG. **4**, aspects of a method **400** for evaluating a resource access request from an Internet-of-things device **116** using a security management service will be described in detail, according to an illustrative embodiment. For purposes of illustrating and describing the concepts of the present disclosure, the method **400** is described herein as being performed by the server computer **102** via execution of one or more software modules such as, for example, the security management service **106**. It should be understood that additional and/or alternative devices and/or network nodes can provide the functionality described herein via execution of one or more modules, applications, and/or other software including, but not limited to, the security management service **106**. Thus, the illustrated embodiments are illustrative, and should not be viewed as being limiting in any way.

(75) The method **400** begins at operation **402**. At operation **402**, the server computer **102** can receive or detect a request (e.g., the resource access request **134**) from an Internet-of-things device **116**. It can be appreciated that the resource access request **134** received in operation **402** can be sent by the Internet-of-things device **116** via the cellular network **124** and received by the server computer **102** via the cellular network **124**. Of course, it must be appreciated that the server computer **102** may not have cellular connectivity in various embodiments, and therefore other networks may be involved in transfer of the resource access request **134** from the Internet-of-things device **116** to the server computer **102**. In various embodiments of operation **402**, however, the resource access request **134** is transmitted from the Internet-of-things device **116** to the server computer **102** via the cellular network **124**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(76) The resource access request **134** received in operation **402** can specify a resource to which the

Internet-of-things device **116** is requesting access. The resource access request **134** can include, in some embodiments, an implicit request. For example, the Internet-of-things device **116** may generate a request to access a particular resource such as a website, a file, an application or the like provided by the third party services **128**, the network infrastructure **130**, and/or the cloud applications **132** in some embodiments. In some other embodiments, the resource access request **134** obtained in operation **402** can include an explicit request that requests access to the resource such as a website, a file, an application or the like provided by the third party services **128**, the network infrastructure **130**, and/or the cloud applications **132**. Thus, operation **402** can include an explicit request generated by the Internet-of-things device **116** in some embodiments, or an implicitly-determined request for access to a resource. Because the request to access a resource can be detected or inferred in various manners, it should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way.

(77) From operation **402**, the method **400** can proceed to operation **404**. At operation **404**, the server computer **102** can analyze identity information to validate the request received or detected in operation **402**. Thus, operation **404** can correspond to the server computer **102** determining the identity of the Internet-of-things device **116** (e.g., via the universal integrated circuit card **120**) and comparing that to other obtained identifying information (e.g., IP addresses, MAC addresses, etc.) to determine if the request obtained in operation **404** was received from the Internet-of-things device **116** (and not any other device spoofing and/or attempting to emulate the Internet-of-things device **116**). It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(78) Operation **404** therefore can correspond to the server computer **102** accessing the device inventory **114**, the identity database **112**, and/or other information to confirm the identity of the Internet-of-things device **116**. Thus, operation **404** can correspond to a continuing analysis of identity associated with the Internet-of-things device **116** to ensure that the Internet-of-things device **116** is still the same device that was authorized for registration with the cellular network **124** previously. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(79) It can be appreciated that the functionality of operation **404** can be similar, in some embodiments, to the analysis performed by the server computer **102** when determining if the Internet-of-things device **116** should be registered with the cellular network **124**, but in operation **404** the analysis can be performed on the specific resource access request **134** received or detected in operation **402** as opposed to being performed on the Internet-of-things device **116** in general. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(80) From operation **404**, the method **400** can proceed to operation **406**. At operation **406**, the server computer **102** can analyze behavior of the Internet-of-things device **116** to identify any threats associated with the access requested in operation **402**. Thus, operation **406** can correspond to the server computer **102** performing a threat evaluation for the Internet-of-things device **116** and/or the access requested by way of the resource access request **134**. In operation **406**, the server computer **102** can analyze the requested access to detect malware signatures, unauthorized access, behavior anomalies, and the like. Thus, if the access or resource requested by way of the resource access request **134** received or detected in operation **402** is not consistent with the Internet-of-things device **116** baseline as determined by the security management service **106**, the server computer **102** can detect such a deviation via the analysis in operation **406**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(81) Operation **404** can correspond to the server computer **102** performing anomaly detection and/or threat detection on the resource access request **134**. Thus, operation **404** can correspond to a continuing analysis of activity associated with the Internet-of-things device **116** to ensure that the Internet-of-things device **116** is still operating normally and not suspect of being under the control

of malware and/or any other unauthorized entity or process, in some embodiments.

(82) It can be appreciated that the functionality of operation **406** can be similar, in some embodiments, to the analysis performed by the server computer **102** when determining if the Internet-of-things device **116** should be registered with the cellular network **124**, but in operation **406**, the analysis can be performed on the specific resource access request **134** received or detected in operation **402** and/or associated behavior of the Internet-of-things device **116** as opposed to being performed on the Internet-of-things device **116** in general. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(83) From operation **406**, the method **400** can proceed to operation **408**. At operation **408**, the server computer **102** can determine if the access requested by way of the resource access request **134** is to be granted. According to various embodiments of the concepts and technologies disclosed herein, the server computer **102** can again obtain and analyze behavioral data (e.g., contextual data) associated with the Internet-of-things device **116**, identity data associated with the Internet-of-things device **116**, data from the identity database **112**, data from the device inventory **114**, and/or other information to determine if the Internet-of-things device **116** is to be allowed to access the resource as requested.

(84) According to various embodiments of the concepts and technologies disclosed herein, if the server computer **102** determines that the identity data (e.g., the unique identity stored in the universal integrated circuit card **120**) associated with the Internet-of-things device **116** is not the same identity that was authorized to register with the cellular network **124** and/or that the Internet-of-things device **116** is not authorized to access the resource as requested in operation **402** via the cellular network **124**; that the corroborating identity information does not match the identity data obtained from the universal integrated circuit card **120** of the Internet-of-things device **116**; that the Internet-of-things device **116** is not operating as expected and/or is operating with one or more anomaly; or that malware or unauthorized control of the Internet-of-things device **116** is detected; the Internet-of-things device **116** can be determined not to be entitled to access the resource via the cellular network **124**, and the method **400** can proceed to operation **410**. At operation **410**, the server computer **102** can block access to the resource requested by way of the resource access request **134**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(85) If the server computer **102** determines, in operation **408**, that the identity data (e.g., the unique identity stored in the universal integrated circuit card **120**) associated with the Internet-of-things device **116** is the same entity that was authorized to register with the cellular network **124** and/or that the Internet-of-things device **116** is authorized to access the resource as requested in operation **402** via the cellular network **124**; that the corroborating identity information matches the identity data obtained from the universal integrated circuit card **120** of the Internet-of-things device **116**; that the Internet-of-things device **116** is operating as expected and without anomalies; and that no malware and/or unauthorized control of the Internet-of-things device **116** is detected; the Internet-of-things device **116** can be determined to be entitled to access the requested resource and/or to obtain the requested access via the cellular network **124**, and the method **400** can proceed to operation **412**.

(86) At operation **412**, the server computer **102** can determine if a certificate **122** is required for the access requested by way of the request received or detected in operation **402**. In operation **412**, the server computer **102** can determine, for example, if the resource to which access is being requested requires a certificate **122** for communications and/or if the resource has requested a certificate **122**. For example, some cloud applications **132** may require the certificate **122** to begin a session with the Internet-of-things device **116** in some embodiments. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(87) If the server computer **102** determines, in operation **412**, that the access requested by way of the resource access request **134** requires a certificate **122**, the method **400** can proceed to operation

414. At operation **414**, the server computer **102** can provide the certificate **122** to the resource. As explained above, the server computer **102** can store certificates **122** for multiple Internet-of-things devices **116** in some embodiments. For example, in some embodiments the server computer **102** can store one or more certificate **122** for one or more Internet-of-things device **116** in the device inventory **114**. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(88) In some other embodiments, the server computer **102** can be configured to obtain certificates **122** from the Internet-of-things device **116** with the resource access request **134**, on-demand or by request, and/or at other times. Thus, the server computer **102** can store and/or can obtain the certificate **122** in operation **414** and provide the certificate **122** to the resource. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(89) From operation **414**, the method **400** can proceed to operation **416**. The method **400** also can proceed to operation **416** if the server computer **102** determines, in operation **412**, that the access requested by way of the resource access request **134** does not require a certificate **122**. At operation **416**, the server computer **102** can determine routing for session data **136** associated with accessing the resource as requested by way of the resource access request **134**.

(90) In operation **416**, the server computer **102** can determine whether the Internet-of-things device **116** should be enabled to communicate directly with the resource (e.g., by exchanging session data **136** with the resource directly via the cellular network **124** and the network **104**), or if the session data **136** is to be routed through the server computer **102** during the session (e.g., the server computer **102** will communicate directly with the resource via the network **104** to exchange session data **136** with the resource, and will communicate with the Internet-of-things device **116** via the cellular network **124** to exchange the session data **136** with the Internet-of-things device **116**).

(91) Thus, it can be appreciated that in some embodiments, the server computer **102** can be configured to set up connections between the Internet-of-things device **116** and the resource (e.g., build a connection that has the Internet-of-things device **116** as a first endpoint, the resource as the second endpoint, and the cellular network **124** and the network **104** as enabling connections), while in some other embodiments, the server computer **102** can function as an intermediary for the Internet-of-things device **116** and the resource.

(92) In some embodiments, the server computer **102** can opt for routing session data **136** through the server computer **102** to enhance security of the security management service **106** (e.g., by performing deep packet inspection (“DPI”) on the session data **136**, by metering the traffic to prevent overflow attacks, man-in-the-middle (“MITM”) attacks, or the like) and/or for other reasons (e.g., load balancing or the like). In some other embodiments, the server computer **102** can opt to enable direct communications between the Internet-of-things device **116** and the resource via the cellular network **124** and the network **104** to provide lower latency and/or higher quality of service (“QoS”) than may be possible if the server computer **102** manages exchange of the session data **136**. Because the routing of the session data **136** can be determined in additional and/or alternative manners, and because the choice of routing can be based on additional and/or alternative considerations, it should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way.

(93) From operation **416**, the method **400** can proceed to operation **418**. At operation **418**, the server computer **102** can grant the access to the resource as requested by way of the resource access request **134**. It can be appreciated that the access granted in operation **418** can be granted in accordance with the routing determined in operation **416**. Thus, the server computer **102** can grant, in operation **418**, direct access to the resource and/or access directed through the server computer **102**. It should be understood that these examples are illustrative, and therefore should not be construed as being limiting in any way.

(94) From operation **418**, the method **400** can proceed to operation **420**. The method **400** can also

proceed to operation **420** from operation **410**. The method **400** can end at operation **420**.

(95) Turning now to FIG. 5, additional details of the network **104** are illustrated, according to an illustrative embodiment. The network **104** includes a cellular network **502** (for example, the cellular network **124** illustrated and described hereinabove), a packet data network **504**, for example, the Internet, and a circuit switched network **506**, for example, a publicly switched telephone network (“PSTN”). The cellular network **502** (and/or the cellular network **124**) includes various components such as, but not limited to, base transceiver stations (“BTSS”), Node-B's or e-Node-B's, base station controllers (“BSCs”), radio network controllers (“RNCs”), mobile switching centers (“MSCs”), mobile management entities (“MMEs”), short message service centers (“SMSCs”), multimedia messaging service centers (“MMSCs”), home location registers (“HLRs”), home subscriber servers (“HSSs”), visitor location registers (“VLRs”), charging platforms, billing platforms, voicemail platforms, GPRS core network components, location service nodes, an IP Multimedia Subsystem (“IMS”), and the like. The cellular network **502** (and/or the cellular network **124**) also includes radios and nodes for receiving and transmitting voice, data, and combinations thereof to and from radio transceivers, networks, the packet data network **504**, and the circuit switched network **506**.

(96) A mobile communications device **508**, such as, for example, a cellular telephone, a user equipment, a mobile terminal, a PDA, a laptop computer, a handheld computer, and combinations thereof, can be operatively connected to the cellular network **502** (and/or the cellular network **124**). The cellular network **502** (and/or the cellular network **124**) can be configured as a 2G GSM network and can provide data communications via GPRS and/or EDGE. Additionally, or alternatively, the cellular network **502** (and/or the cellular network **124**) can be configured as a 3G UMTS network and can provide data communications via the HSPA protocol family, for example, HSDPA, EUL (also referred to as HSUPA), and HSPA+. The cellular network **502** (and/or the cellular network **124**) also is compatible with 4G mobile communications standards, 5G mobile communications standards, other mobile communications standards, and evolved and future mobile communications standards.

(97) The packet data network **504** includes various devices, for example, servers, computers, databases, and other devices in communication with one another, as is generally known. The packet data network **504** devices are accessible via one or more network links. The servers often store various files that are provided to a requesting device such as, for example, a computer, a terminal, a smartphone, or the like. Typically, the requesting device includes software (a “browser”) for executing a web page in a format readable by the browser or other software. Other files and/or data may be accessible via “links” in the retrieved files, as is generally known. In some embodiments, the packet data network **504** includes or is in communication with the Internet. The circuit switched network **506** includes various hardware and software for providing circuit switched communications. The circuit switched network **506** may include, or may be, what is often referred to as a plain old telephone system (POTS). The functionality of a circuit switched network **506** or other circuit-switched network are generally known and will not be described herein in detail.

(98) The illustrated cellular network **502** (and/or the cellular network **124**) is shown in communication with the packet data network **504** and a circuit switched network **506**, though it should be appreciated that this is not necessarily the case. One or more Internet-capable devices **510**, for example, a PC, a laptop, a portable device, or another suitable device, can communicate with one or more cellular networks **502** (and/or the cellular network **124**), and devices connected thereto, through the packet data network **504**. It also should be appreciated that the Internet-capable device **510** can communicate with the packet data network **504** through the circuit switched network **506**, the cellular network **502** (and/or the cellular network **124**), and/or via other networks (not illustrated).

(99) As illustrated, a communications device **512**, for example, a telephone, facsimile machine, modem, computer, or the like, can be in communication with the circuit switched network **506**, and

therethrough to the packet data network **504** and/or the cellular network **502** (and/or the cellular network **124**). It should be appreciated that the communications device **512** can be an Internet-capable device, and can be substantially similar to the Internet-capable device **510**. In the specification, the network **104** is used to refer broadly to any combination of the networks **502**, **504**, **506** (and/or the cellular network **124**). It should be appreciated that substantially all of the functionality described with reference to the network **104** can be performed by the cellular network **502** (and/or the cellular network **124**), the packet data network **504**, and/or the circuit switched network **506**, alone or in combination with other networks, network elements, and the like.

(100) FIG. **6** is a block diagram illustrating a computer system **600** configured to provide the functionality described herein for providing and using a security management service for Internet-of-things devices, in accordance with various embodiments of the concepts and technologies disclosed herein. The computer system **600** includes a processing unit **602**, a memory **604**, one or more user interface devices **606**, one or more input/output (“I/O”) devices **608**, and one or more network devices **610**, each of which is operatively connected to a system bus **612**. The bus **612** enables bi-directional communication between the processing unit **602**, the memory **604**, the user interface devices **606**, the I/O devices **608**, and the network devices **610**.

(101) The processing unit **602** may be a standard central processor that performs arithmetic and logical operations, a more specific purpose programmable logic controller (“PLC”), a programmable gate array, or other type of processor known to those skilled in the art and suitable for controlling the operation of the server computer. As used herein, the word “processor” and/or the phrase “processing unit” when used with regard to any architecture or system can include multiple processors or processing units distributed across and/or operating in parallel in a single machine or in multiple machines. Furthermore, processors and/or processing units can be used to support virtual processing environments. Processors and processing units also can include state machines, application-specific integrated circuits (“ASICs”), combinations thereof, or the like. Because processors and/or processing units are generally known, the processors and processing units disclosed herein will not be described in further detail herein.

(102) The memory **604** communicates with the processing unit **602** via the system bus **612**. In some embodiments, the memory **604** is operatively connected to a memory controller (not shown) that enables communication with the processing unit **602** via the system bus **612**. The memory **604** includes an operating system **614** and one or more program modules **616**. The operating system **614** can include, but is not limited to, members of the WINDOWS, WINDOWS CE, and/or WINDOWS MOBILE families of operating systems from MICROSOFT CORPORATION, the LINUX family of operating systems, the SYMBIAN family of operating systems from SYMBIAN LIMITED, the BREW family of operating systems from QUALCOMM CORPORATION, the MAC OS, iOS, and/or LEOPARD families of operating systems from APPLE CORPORATION, the FREEBSD family of operating systems, the SOLARIS family of operating systems from ORACLE CORPORATION, other operating systems, and the like.

(103) The program modules **616** may include various software and/or program modules described herein. In some embodiments, for example, the program modules **616** include the security management service **106**, the credential manager **108**, the routing controller **110**, the security application **118**, the third party services **128**, the cloud applications **132**, and/or one or more components of the network infrastructure **130**. These and/or other programs can be embodied in computer-readable media containing instructions that, when executed by the processing unit **602**, perform one or more of the methods **200**, **300**, and **400** described in detail above with respect to FIGS. **2-4** and/or other functionality as illustrated and described herein. It can be appreciated that, at least by virtue of the instructions embodying the methods **200**, **300**, and **400**, and/or other functionality illustrated and described herein being stored in the memory **604** and/or accessed and/or executed by the processing unit **602**, the computer system **600** is a special-purpose computing system that can facilitate providing the functionality illustrated and described herein.

According to embodiments, the program modules **616** may be embodied in hardware, software, firmware, or any combination thereof. Although not shown in FIG. 6, it should be understood that the memory **604** also can be configured to store the identity database **112**, the device inventory **114**, the universal integrated circuit card **120**, the certificate **122**, the registration data **126**, the resource access request **134**, the session data **136**, and/or other data, if desired.

(104) By way of example, and not limitation, computer-readable media may include any available computer storage media or communication media that can be accessed by the computer system **600**. Communication media includes computer-readable instructions, data structures, program modules, or other data in a modulated data signal such as a carrier wave or other transport mechanism and includes any delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics changed or set in a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.

(105) Computer storage media includes only non-transitory embodiments of computer readable media as illustrated and described herein. Thus, computer storage media can include volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. Computer storage media includes, but is not limited to, RAM, ROM, Erasable Programmable ROM (“EPROM”), Electrically Erasable Programmable ROM (“EEPROM”), flash memory or other solid state memory technology, CD-ROM, digital versatile disks (“DVD”), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computer system **600**. In the claims, the phrase “computer storage medium” and variations thereof does not include waves or signals per se and/or communication media.

(106) The user interface devices **606** may include one or more devices with which a user accesses the computer system **600**. The user interface devices **606** may include, but are not limited to, computers, servers, personal digital assistants, cellular phones, or any suitable computing devices. The I/O devices **608** enable a user to interface with the program modules **616**. In one embodiment, the I/O devices **608** are operatively connected to an I/O controller (not shown) that enables communication with the processing unit **602** via the system bus **612**. The I/O devices **608** may include one or more input devices, such as, but not limited to, a keyboard, a mouse, or an electronic stylus. Further, the I/O devices **608** may include one or more output devices, such as, but not limited to, a display screen or a printer.

(107) The network devices **610** enable the computer system **600** to communicate with other networks or remote systems via a network, such as the network **104** and/or the cellular network **124**. Examples of the network devices **610** include, but are not limited to, a modem, a radio frequency (“RF”) or infrared (“IR”) transceiver, a telephonic interface, a bridge, a router, or a network card. The network **104** may include a wireless network such as, but not limited to, a Wireless Local Area Network (“WLAN”) such as a WI-FI network, a Wireless Wide Area Network (“WWAN”), a Wireless Personal Area Network (“WPAN”) such as BLUETOOTH, a Wireless Metropolitan Area Network (“WMAN”) such as a WiMAX network, or a cellular network. Alternatively, the network **104** may be a wired network such as, but not limited to, a Wide Area Network (“WAN”) such as the Internet, a Local Area Network (“LAN”) such as the Ethernet, a wired Personal Area Network (“PAN”), or a wired Metropolitan Area Network (“MAN”).

(108) Turning now to FIG. 7, an illustrative Internet-of-things device **116** and components thereof will be described. It should be understood that not all embodiments of the Internet-of-things device **116** include all of the components and/or functionality described herein with reference to FIG. 7. While connections are not shown between the various components illustrated in FIG. 7, it should

be understood that some, none, or all of the components illustrated in FIG. 7 can be configured to interact with one another to carry out various device functions. In some embodiments, the components are arranged so as to communicate via one or more busses (not shown). Thus, it should be understood that FIG. 7 and the following description are intended to provide a general understanding of a suitable environment in which various aspects of embodiments can be implemented, and should not be construed as being limiting in any way.

(109) As illustrated in FIG. 7, the Internet-of-things device **116** can include a display **702** for displaying various types of information and/or data. According to various embodiments, the display **702** can be configured to display various graphical user interface (“GUI”) elements such as, for example, connection information, identity information, text, images, video, virtual keypads and/or keyboards, messaging data, notification messages, metadata, internet content, device status, time, date, calendar data, device preferences, map and location data, combinations thereof, and/or the like. The Internet-of-things device **116** also can include a processor **704** and a memory or other data storage device (“memory”) **706**. The processor **704** can be configured to process data and/or can execute computer-executable instructions stored in the memory **706**. The computer-executable instructions executed by the processor **704** can include, for example, an operating system **708**, one or more applications **710** such as the security application **118** and/or other computer-executable instructions stored in a memory **706**, or the like.

(110) In some embodiments, the applications **710** also can include a UI application (not illustrated in FIG. 7). If included, the UI application can interface with the operating system **708** to facilitate user interaction with functionality and/or data stored at the Internet-of-things device **116** and/or stored elsewhere. In some embodiments, the operating system **708** can include various members of various families of operating systems, as is generally understood. The UI application can be executed by the processor **704** to aid a user in entering content, identity information, storing certificates **122**, managing authentication information, configuring settings, interacting with other applications **710**, and otherwise facilitating user interaction with the operating system **708**, the applications **710**, and/or other types or instances of data **712** that can be stored at the Internet-of-things device **116**. The data **712** can include, for example, applications or program modules such as, for example, presence applications, visual voice mail applications, messaging applications, text-to-speech and speech-to-text applications, add-ons, plug-ins, email applications, music applications, video applications, camera applications, location-based service applications, power conservation applications, game applications, productivity applications, authentication applications, identity-management applications, entertainment applications, enterprise applications, combinations thereof, and the like. The applications **710**, the data **712**, and/or portions thereof can be stored in the memory **706** and/or in a firmware **714**, and can be executed by the processor **704**.

(111) It can be appreciated that, at least by virtue of storage of the instructions corresponding to the applications **710** and/or other instructions embodying other functionality illustrated and described herein in the memory **706**, and/or by virtue of the instructions corresponding to the applications **710** and/or other instructions embodying other functionality illustrated and described herein being accessed and/or executed by the processor **704**, the Internet-of-things device **116** is a special-purpose device that can facilitate providing the functionality illustrated and described herein. The firmware **714** also can store code for execution during device power up and power down operations. It can be appreciated that the firmware **714** can be stored in a volatile or non-volatile data storage device including, but not limited to, the memory **706** and/or a portion thereof.

(112) The Internet-of-things device **116** also can include an input/output (“I/O”) interface **716**. The I/O interface **716** can be configured to support the input/output of data such as location information, storing certificates **122**, flashing the universal integrated circuit card **120**, storing identity modules, user information, organization information, presence status information, user IDs, passwords, and application initiation (start-up) requests. In some embodiments, the I/O interface **716** can include a hardwire connection such as a universal serial bus (“USB”) port, a mini-USB

port, a micro-USB port, an audio jack, a PS2 port, an IEEE 1394 (“FIREWIRE”) port, a serial port, a parallel port, an Ethernet port, a telephone port, a proprietary port, combinations thereof, or the like. In some embodiments, the I/O interface **716** accepts I/O devices such as keyboards, keypads, mice, interface tethers, printers, plotters, external storage, touch/multi-touch screens, touch pads, trackballs, joysticks, microphones, remote control devices, displays, projectors, medical equipment (e.g., stethoscopes, heart monitors, and other health metric monitors), modems, routers, external power sources, docking stations, combinations thereof, and the like. It should be appreciated that the I/O interface **716** may be used for communications between the Internet-of-things device **116** and a network device or local device.

(113) The Internet-of-things device **116** also can include a communications component **718**. The communications component **718** can be configured to interface with the processor **704** to facilitate wireless communications with one or more networks such as the cellular network **124** and/or the network **104** described herein. In some embodiments, other networks include networks that utilize non-cellular wireless technologies such as WI-FI or WIMAX. In some embodiments, the communications component **718** includes a multimode communications subsystem for facilitating communications via the cellular network and one or more other networks.

(114) The communications component **718** can include one or more transceivers. The one or more transceivers can be configured to support communications with the cellular network **124** via the same and/or different wireless technology standards with respect to one another. For example, in some embodiments one or more of the transceivers of the communications component **718** may be configured to communicate using GSM, CDMAONE, CDMA2000, LTE, and various other 2G, 2.5G, 3G, 4G, 5G, and greater generation technology standards. Moreover, the communications component **718** may facilitate communications over various channel access methods (which may or may not be used by the aforementioned standards) including, but not limited to, TDMA, FDMA, W-CDMA, OFDM, SDMA, and the like.

(115) In addition, the communications component **718** may facilitate data communications using GPRS, EDGE, the HSPA protocol family including HSDPA, EUL or otherwise termed HSUPA, HSPA+, and various other current and future wireless data access standards. In the illustrated embodiment, the communications component **718** can include a first transceiver (“TxRx”) **720A** that can operate in a first communications mode (e.g., GSM). The communications component **718** also can include an N.sup.th transceiver (“TxRx”) **720N** that can operate in a second communications mode relative to the first transceiver **720A** (e.g., UMTS). While two transceivers **720A-N** (hereinafter collectively and/or generically referred to as “transceivers **720**”) are shown in FIG. 7, it should be appreciated that less than two, two, and/or more than two transceivers **720** can be included in the communications component **718**.

(116) The communications component **718** also can include an alternative transceiver (“Alt TxRx”) **722** for supporting other types and/or standards of communications. According to various contemplated embodiments, the alternative transceiver **722** can communicate using various communications technologies such as, for example, WI-FI, WIMAX, BLUETOOTH, infrared, infrared data association (“IRDA”), near field communications (“NFC”), other RF technologies, combinations thereof, and the like. In some embodiments, the communications component **718** also can facilitate reception from terrestrial radio networks, digital satellite radio networks, internet-based radio service networks, combinations thereof, and the like. The communications component **718** can process data from a network such as the Internet, an intranet, a broadband network, a WI-FI hotspot, an Internet service provider (“ISP”), a digital subscriber line (“DSL”) provider, a broadband provider, combinations thereof, or the like.

(117) The Internet-of-things device **116** also can include one or more sensors **724**. The sensors **724** can include temperature sensors, light sensors, air quality sensors, movement sensors, orientation sensors, noise sensors, proximity sensors, or the like. As such, it should be understood that the sensors **724** can include, but are not limited to, accelerometers, magnetometers, gyroscopes,

infrared sensors, noise sensors, microphones, combinations thereof, or the like. Additionally, audio capabilities for the Internet-of-things device **116** may be provided by an audio I/O component **726**. The audio I/O component **726** of the Internet-of-things device **116** can include one or more speakers for the output of audio signals, one or more microphones for the collection and/or input of audio signals, and/or other audio input and/or output devices.

(118) The illustrated Internet-of-things device **116** also can include a SIM system **728**. The SIM system **728** can include a universal SIM (“USIM”), a universal integrated circuit card such as the universal integrated circuit card **120** illustrated and described herein, and/or other identity devices. In various embodiments, the SIM system **728** can be provided by a software module. In some embodiments, the SIM system **728** also can store one or more certificates **122** as illustrated and described herein. In some embodiments, the SIM system **728** can include and/or can be connected to or inserted into an interface such as a slot interface **730**. In some embodiments, the slot interface **730** can be configured to accept insertion of other identity cards or modules for accessing various types of networks. Additionally, or alternatively, the slot interface **730** can be configured to accept multiple subscriber identity cards. Because other devices and/or modules for identifying users and/or the Internet-of-things device **116** are contemplated, it should be understood that these embodiments are illustrative, and should not be construed as being limiting in any way.

(119) The Internet-of-things device **116** also can include an image capture and processing system **732** (“image system”). The image system **732** can be configured to capture or otherwise obtain photos, videos, and/or other visual information. As such, the image system **732** can include cameras, lenses, charge-coupled devices (“CCDs”), combinations thereof, or the like. The Internet-of-things device **116** may also include a video system **734**. The video system **734** can be configured to capture, process, record, modify, and/or store video content. Photos and videos obtained using the image system **732** and the video system **734**, respectively, may be added as message content to an MMS message, email message, and sent to another mobile device. The video and/or photo content also can be shared with other devices via various types of data transfers via wired and/or wireless communication devices as described herein.

(120) The Internet-of-things device **116** also can include one or more location components **736**. The location components **736** can be configured to send and/or receive signals to determine a geographic location of the Internet-of-things device **116**. According to various embodiments, the location components **736** can send and/or receive signals from global positioning system (“GPS”) devices, assisted-GPS (“A-GPS”) devices, WI-FI/WIMAX and/or cellular network triangulation data, combinations thereof, and the like. The location component **736** also can be configured to communicate with the communications component **718** to retrieve triangulation data for determining a location of the Internet-of-things device **116**. In some embodiments, the location component **736** can interface with cellular network nodes, telephone lines, satellites, location transmitters and/or beacons, wireless network transmitters and receivers, combinations thereof, and the like. In some embodiments, the location component **736** can include and/or can communicate with one or more of the sensors **724** such as a compass, an accelerometer, and/or a gyroscope to determine the orientation of the Internet-of-things device **116**. Using the location component **736**, the Internet-of-things device **116** can generate and/or receive data to identify its geographic location, or to transmit data used by other devices to determine the location of the Internet-of-things device **116**. The location component **736** may include multiple components for determining the location and/or orientation of the Internet-of-things device **116**.

(121) The illustrated Internet-of-things device **116** also can include a power source **738**. The power source **738** can include one or more batteries, power supplies, power cells, and/or other power subsystems including alternating current (“AC”) and/or direct current (“DC”) power devices. The power source **738** also can interface with an external power system or charging equipment via a power I/O component **740**. Because the Internet-of-things device **116** can include additional and/or alternative components, the above embodiment should be understood as being illustrative of one

possible operating environment for various embodiments of the concepts and technologies described herein. The described embodiment of the Internet-of-things device **116** is illustrative, and should not be construed as being limiting in any way.

(122) FIG. **8** illustrates an illustrative architecture for a cloud computing platform **800** that can be capable of executing the software components described herein and/or for interacting with the security management service **106**. Thus, it can be appreciated that in some embodiments of the concepts and technologies disclosed herein, the cloud computing platform **800** illustrated in FIG. **8** can be used to provide the functionality described herein with respect to the server computer **102** and/or one or more components of the network infrastructure **130**.

(123) The cloud computing platform **800** thus may be utilized to execute any aspects of the software components presented herein. Thus, according to various embodiments of the concepts and technologies disclosed herein, the security management service **106**, the credential manager **108**, the routing controller **110**, and/or one or more components of the network infrastructure **130** can be implemented, at least in part, on or by elements included in the cloud computing platform **800** illustrated and described herein. Those skilled in the art will appreciate that the illustrated cloud computing platform **800** is a simplification of but only one possible implementation of an illustrative cloud computing platform, and as such, the illustrated cloud computing platform **800** should not be construed as being limiting in any way.

(124) In the illustrated embodiment, the cloud computing platform **800** can include a hardware resource layer **802**, a virtualization/control layer **804**, and a virtual resource layer **806**. These layers and/or other layers can be configured to cooperate with each other and/or other elements of a cloud computing platform **800** to perform operations as will be described in detail herein. While connections are shown between some of the components illustrated in FIG. **8**, it should be understood that some, none, or all of the components illustrated in FIG. **8** can be configured to interact with one another to carry out various functions described herein. In some embodiments, the components are arranged so as to communicate via one or more networks such as, for example, the network **104** illustrated and described hereinabove (not shown in FIG. **8**). Thus, it should be understood that FIG. **8** and the following description are intended to provide a general understanding of a suitable environment in which various aspects of embodiments can be implemented, and should not be construed as being limiting in any way.

(125) The hardware resource layer **802** can provide hardware resources. In the illustrated embodiment, the hardware resources can include one or more compute resources **808**, one or more memory resources **810**, and one or more other resources **812**. The compute resource(s) **808** can include one or more hardware components that can perform computations to process data, and/or to execute computer-executable instructions of one or more application programs, operating systems, services, and/or other software including, but not limited to, the security management service **106**, the credential manager **108**, the routing controller **110**, and/or one or more components of the network infrastructure **130** illustrated and described herein.

(126) According to various embodiments, the compute resources **808** can include one or more central processing units (“CPUs”). The CPUs can be configured with one or more processing cores. In some embodiments, the compute resources **808** can include one or more graphics processing units (“GPUs”). The GPUs can be configured to accelerate operations performed by one or more CPUs, and/or to perform computations to process data, and/or to execute computer-executable instructions of one or more application programs, operating systems, and/or other software that may or may not include instructions that are specifically graphics computations and/or related to graphics computations. In some embodiments, the compute resources **808** can include one or more discrete GPUs. In some other embodiments, the compute resources **808** can include one or more CPU and/or GPU components that can be configured in accordance with a co-processing CPU/GPU computing model. Thus, it can be appreciated that in some embodiments of the compute resources **808**, a sequential part of an application can execute on a CPU and a computationally-

intensive part of the application can be accelerated by the GPU. It should be understood that this example is illustrative, and therefore should not be construed as being limiting in any way.

(127) In some embodiments, the compute resources **808** also can include one or more system on a chip (“SoC”) components. It should be understood that an SoC component can operate in association with one or more other components as illustrated and described herein, for example, one or more of the memory resources **810** and/or one or more of the other resources **812**. In some embodiments in which an SoC component is included, the compute resources **808** can be or can include one or more embodiments of the SNAPDRAGON brand family of SoCs, available from QUALCOMM of San Diego, California; one or more embodiment of the TEGRA brand family of SoCs, available from NVIDIA of Santa Clara, California; one or more embodiment of the HUMMINGBIRD brand family of SoCs, available from SAMSUNG of Seoul, South Korea; one or more embodiment of the Open Multimedia Application Platform (“OMAP”) family of SoCs, available from TEXAS INSTRUMENTS of Dallas, Texas; one or more customized versions of any of the above SoCs; and/or one or more other brand and/or one or more proprietary SoCs.

(128) The compute resources **808** can be or can include one or more hardware components arranged in accordance with an ARM architecture, available for license from ARM HOLDINGS of Cambridge, United Kingdom. Alternatively, the compute resources **808** can be or can include one or more hardware components arranged in accordance with an x86 architecture, such as an architecture available from INTEL CORPORATION of Mountain View, California, and others. Those skilled in the art will appreciate the implementation of the compute resources **808** can utilize various computation architectures and/or processing architectures. As such, the various example embodiments of the compute resources **808** as mentioned hereinabove should not be construed as being limiting in any way. Rather, implementations of embodiments of the concepts and technologies disclosed herein can be implemented using compute resources **808** having any of the particular computation architecture and/or combination of computation architectures mentioned herein as well as other architectures.

(129) Although not separately illustrated in FIG. **8**, it should be understood that the compute resources **808** illustrated and described herein can host and/or execute various services, applications, portals, and/or other functionality illustrated and described herein. Thus, the compute resources **808** can host and/or can execute the security management service **106**, the credential manager **108**, the routing controller **110**, and/or one or more components of the network infrastructure **130** or other applications or services illustrated and described herein.

(130) The memory resource(s) **810** can include one or more hardware components that can perform or provide storage operations, including temporary and/or permanent storage operations. In some embodiments, the memory resource(s) **810** can include volatile and/or non-volatile memory implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data disclosed herein. Computer storage media is defined hereinabove and therefore should be understood as including, in various embodiments, random access memory (“RAM”), read-only memory (“ROM”), Erasable Programmable ROM (“EPROM”), Electrically Erasable Programmable ROM (“EEPROM”), flash memory or other solid state memory technology, CD-ROM, digital versatile disks (“DVD”), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store data and that can be accessed by the compute resources **808**, subject to the definition of “computer storage media” provided above (e.g., as excluding waves and signals per se and/or communication media as defined in this application).

(131) Although not illustrated in FIG. **8**, it should be understood that the memory resources **810** can host or store the various data illustrated and described herein including, but not limited to, the identity database **112**, the device inventory **114**, the universal integrated circuit card **120**, the certificate **122**, the registration data **126**, the resource access request **134**, the session data **136**, and/or other data, if desired. It should be understood that this example is illustrative, and therefore

should not be construed as being limiting in any way.

(132) The other resource(s) **812** can include any other hardware resources that can be utilized by the compute resources(s) **808** and/or the memory resource(s) **810** to perform operations. The other resource(s) **812** can include one or more input and/or output processors (e.g., a network interface controller and/or a wireless radio), one or more modems, one or more codec chipsets, one or more pipeline processors, one or more fast Fourier transform (“FFT”) processors, one or more digital signal processors (“DSPs”), one or more speech synthesizers, combinations thereof, or the like.

(133) The hardware resources operating within the hardware resource layer **802** can be virtualized by one or more virtual machine monitors (“VMMs”) **814A-814N** (also known as “hypervisors;” hereinafter “VMMs **814**”). The VMMs **814** can operate within the virtualization/control layer **804** to manage one or more virtual resources that can reside in the virtual resource layer **806**. The VMMs **814** can be or can include software, firmware, and/or hardware that alone or in combination with other software, firmware, and/or hardware, can manage one or more virtual resources operating within the virtual resource layer **806**.

(134) The virtual resources operating within the virtual resource layer **806** can include abstractions of at least a portion of the compute resources **808**, the memory resources **810**, the other resources **812**, or any combination thereof. These abstractions are referred to herein as virtual machines (“VMs”). In the illustrated embodiment, the virtual resource layer **806** includes VMs **816A-816N** (hereinafter “VMs **816**”).

(135) Based on the foregoing, it should be appreciated that systems and methods for providing and using a security management service for Internet-of-things devices have been disclosed herein. Although the subject matter presented herein has been described in language specific to computer structural features, methodological and transformative acts, specific computing machinery, and computer-readable media, it is to be understood that the concepts and technologies disclosed herein are not necessarily limited to the specific features, acts, or media described herein. Rather, the specific features, acts and mediums are disclosed as example forms of implementing the concepts and technologies disclosed herein.

(136) The subject matter described above is provided by way of illustration only and should not be construed as limiting. Various modifications and changes may be made to the subject matter described herein without following the example embodiments and applications illustrated and described, and without departing from the true spirit and scope of the embodiments of the concepts and technologies disclosed herein.

Claims

1. A system comprising: a processor; and a memory that stores computer-executable instructions that, when executed by the processor, cause the processor to perform operations comprising obtaining, from an Internet-of-things device and via a portion of a cellular network, registration data, wherein obtaining the registration data corresponds to a request by the Internet-of-things device to register with the cellular network, wherein the Internet-of-things device comprises a cellular transceiver and a universal integrated circuit card that stores a unique identifier for the Internet-of-things device, obtaining, from the registration data, identity data that comprises the unique identifier for the Internet-of-things device, in response to determining, based on the identity data and behavioral data that describes activity associated with the Internet-of-things device, that the Internet-of-things device is operating normally, allowing the Internet-of-things device to register with the cellular network, receiving, from the Internet-of-things device and via the cellular network, a request for the Internet-of-things device to access a resource via another network, obtaining, from an identity database, corroborating identify information that identifies the Internet-of-things device, analyzing the behavioral data, the identity data, and the corroborating identity information to determine if the access requested is to be granted, wherein the access is to be

granted if the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under control of any unauthorized entity, and in response to determining that the access requested is to be granted, allowing the Internet-of-things device to access the resource via the cellular network.

2. The system of claim 1, wherein the corroborating identity information comprises an address associated with the Internet-of-things device, the address comprising an IP address or a MAC address.

3. The system of claim 1, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising: determining if the resource requires a certificate associated with the Internet-of-things device to allow the Internet-of-things device to access the resource; and if a determination is made that the resource requires the certificate, providing the certificate to the resource, wherein the certificate is obtained with the registration data and from the universal integrated circuit card.

4. The system of claim 1, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising: determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing comprising a direct connection between the Internet-of-things device and the resource via the cellular network and the other network.

5. The system of claim 1, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising: determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing comprising an indirect connection between the Internet-of-things device and the resource, the indirect connection comprising communications via the cellular network, a server computer that executes a security management service, and the other network.

6. The system of claim 1, wherein the resource comprises a cloud application.

7. The system of claim 1, wherein the Internet-of-things device stores a certificate in the universal integrated circuit card at the Internet-of-things device, wherein the certificate is stored in a secure memory of the universal integrated circuit card.

8. A method comprising: obtaining, at a server computer comprising a processor and from an Internet-of-things device and via a portion of a cellular network, registration data, wherein obtaining the registration data corresponds to a request by the Internet-of-things device to register with the cellular network, wherein the Internet-of-things device comprises a cellular transceiver and a universal integrated circuit card that stores a unique identifier for the Internet-of-things device; obtaining, by the processor and from the registration data, identity data that comprises the unique identifier for the Internet-of-things device; in response to determining, by the processor and based on the identity data and behavioral data that describes activity associated with the Internet-of-things device, that the Internet-of-things device is operating normally, allowing the Internet-of-things device to register with the cellular network; receiving, by the processor and from the Internet-of-things device and via the cellular network, a request for the Internet-of-things device to access a resource via another network; obtaining, by the processor and from an identity database, corroborating identity information that identifies the Internet-of-things device; analyzing, by the processor, the behavioral data, the identity data, and the corroborating identity information to determine if the access requested is to be granted, wherein the access is to be granted if the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under control of any unauthorized entity; and in response to determining that the access requested is to be granted, allowing, by the processor, the Internet-of-things device to access the resource via the cellular network.

9. The method of claim 8 wherein the corroborating identity information comprises an address associated with the Internet-of-things device, the address comprising an IP address or a MAC address.

10. The method of claim 8, further comprising: determining if the resource requires a certificate associated with the Internet-of-things device to allow the Internet-of-things device to access the resource; and if a determination is made that the resource requires the certificate, providing the certificate to the resource, wherein the certificate is obtained with the registration data and from the universal integrated circuit card.
11. The method of claim 8, further comprising: determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing comprising a direct connection between the Internet-of-things device and the resource via the cellular network and the other network.
12. The method of claim 8, further comprising: determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing comprising an indirect connection between the Internet-of-things device and the resource, the indirect connection comprising communications via the cellular network, the server computer, and the other network.
13. The method of claim 8, wherein the Internet-of-things device stores a certificate in the universal integrated circuit card at the Internet-of-things device, wherein the certificate is stored in a secure memory of the universal integrated circuit card.
14. A computer storage medium having computer-executable instructions stored thereon that, when executed by a processor, cause the processor to perform operations comprising: obtaining, from an Internet-of-things device and via a portion of a cellular network, registration data, wherein obtaining the registration data corresponds to a request by the Internet-of-things device to register with the cellular network, wherein the Internet-of-things device comprises a cellular transceiver and a universal integrated circuit card that stores a unique identifier for the Internet-of-things device; obtaining, from the registration data, identity data that comprises the unique identifier for the Internet-of-things device; in response to determining, based on the identity data and behavioral data that describes activity associated with the Internet-of-things device, that the Internet-of-things device is operating normally, allowing the Internet-of-things device to register with the cellular network; receiving, from the Internet-of-things device and via the cellular network, a request for the Internet-of-things device to access a resource via another network; obtaining, from an identity database, corroborating identity information that identifies the Internet-of-things device; analyzing the behavioral data, the identity data, and the corroborating identity information to determine if the access requested is to be granted, wherein the access is to be granted if the Internet-of-things device is operating normally and as expected and that the Internet-of-things device is not under control of any unauthorized entity; and in response to determining that the access requested is to be granted, allowing the Internet-of-things device to access the resource via the cellular network.
15. The computer storage medium of claim 14, wherein the corroborating identity information comprises a data network name, a location area code, or a tracking area code.
16. The computer storage medium of claim 14, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising: determining if the resource requires a certificate associated with the Internet-of-things device to allow the Internet-of-things device to access the resource; and if a determination is made that the resource requires the certificate, providing the certificate to the resource, wherein the certificate is obtained with the registration data and from the universal integrated circuit card.
17. The computer storage medium of claim 14, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising: determining a routing for session data exchanged between the Internet-of-things device and the resource during a session, the routing comprising a direct connection between the Internet-of-things device and the resource via the cellular network and the other network.
18. The computer storage medium of claim 14, wherein the computer-executable instructions, when executed by the processor, cause the processor to perform operations further comprising: determining a routing for session data exchanged between the Internet-of-things device and the

resource during a session, the routing comprising an indirect connection between the Internet-of-things device and the resource, the indirect connection comprising communications via the cellular network, a server computer that executes a security management service, and the other network.

19. The computer storage medium of claim 14, wherein the resource comprises network infrastructure associated with the other network.

20. The computer storage medium of claim 14, wherein the Internet-of-things device stores a certificate in the universal integrated circuit card at the Internet-of-things device, wherein the certificate is stored in a secure memory of the universal integrated circuit card.
